CHAGEE

CHAGEE SINGAPORE

FORECAST THE PEAK, SHARE THE COUNTER

PREDICTIVE AND PHYSICAL AI: WHERE TO OPEN NEXT, AND WHAT THE GM DECIDES FOR THE BUSIEST HOUR

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SPEAKER NOTES · TITLE

Good morning. We are Group 3. Our company is Chagee, and we look at one place in its business: the counter of a Singapore outlet in its busiest hour.

Two kinds of technology meet there. Predictive AI forecasts demand: where to open, and how busy the hour will be. Physical AI, automation and machines, does the fixed steps. Around them, people share the hand steps like a bucket brigade.

Our decision-maker throughout is the GM, Chagee Singapore's general manager. Timing: title and agenda together, about twenty seconds.

Agenda

01 **Chagee in Singapore**
How it earns and sells

02 **From farm to cup**
Where the supply chain is stable, and where it is not

03 **The data trail**
What every cup already records

04 **Our scope**
Two areas, two kinds of AI, one decision-maker

05 **Predict demand**
App searches, DBSCAN and a new-store pin

06 **Share the counter**
The bucket brigade, side by side

07 **Split the counter**
Which steps stay human: the GM's call

08 **The GM's playbook**
Triggers, pilot and limits

09 **The size of the benefit**
Orders kept, revenue, and what it rests on



SPEAKER NOTES · AGENDA

Eight short chapters. The first four set the scope; the last four are the forecast, the counter, the GM's split and the playbook. Click any chapter to jump there.

In Singapore, outlets run under one company, and every channel ends at one counter

One company and one data system: the counter, the app data and the call on who works the peak sit together (informal source, to confirm).

HOW A SINGAPORE CUP IS SOLD · STEP THROUGH

GROUP NET REVENUE, FY 2025 (SECONDARY SOURCE)



Franchise **88%** Company-owned **12%**, growing fastest

SG Singapore outlets run under one company and one data system (informal source, to confirm with Chagee)

In store	walk-in, counter or kiosk
Chagee app	order ahead, collect
Grab	third-party delivery

→ **One counter** every channel's cup is made here



44 outlets, ground-floor mall units by preference; kiosks in basements and MRT stations at entry in 2019

- **One operator, one data system**
  Singapore outlets run under one company and one data system, so a counter design or roster rule could be rolled out across outlets.
  A Chagee contact (informal), to confirm with Chagee
- **App-first**
  Most orders come through the app, with few kiosks in store. Every order and search is logged: the raw material for a forecast.
  Team observation, to confirm
- **Why Chagee**
  A fast-growing, playful brand with heavy app use: plenty of data, and room for AI to help the stores.

TAKEAWAY | Whatever the channel, every order ends at one counter. The peak hour is where an outlet wins or loses.

SPEAKER NOTES · CHAGEE IN SINGAPORE

Step through four points. One: at group level most revenue comes from franchising, about 88% in FY 2025; company-owned is about 12% and growing fastest. That split is from a secondary source.

Two: Singapore outlets run under one company and one data system. We have this informally from a Chagee contact, who told us the ownership form, owned, joint venture or franchise, does not matter for this question. We will confirm it with Chagee, and we make no claim about the ownership structure.

Three: 44 outlets, by the store directory in October. Chagee prefers ground-floor mall units.

Four: walk-in, app orders for collection, and Grab delivery all end at the same counter. The Chagee app is for ordering ahead and collecting; delivery goes through third-party apps, by the same informal source. That is why we look at the counter. About fifty seconds.

Supply is standard and planned ahead; the counter feels demand within minutes

Tea leaves travel one standard route. Only fresh milk and fruit are bought locally.

FOLLOW ONE CUP · STEP THROUGH



Source

Tea leaves from Yunnan through standard channels for every market

DATA CREATED

Shipments, lead times

Local: fresh milk, fruit

Warehouse

Separate central and regional warehouses keep storage and logistics costs down

DATA CREATED

Stock vs par, expiry

Outlet counter

Machines brew each cup from a scanned recipe; people do ice, toppings and hand-over

DATA CREATED

Cups by hour

Wait times

Last mile

Self-collect from the app, or delivery via third-party apps such as Grab; riders wait at the same counter

DATA CREATED

Orders by channel

Rider wait

Customer

Orders in the app, collects at the counter, comes back (or not)

DATA CREATED

App orders, drop-offs

- **Stable upstream**
  Standard sourcing, regional warehouses and forecast-led restocking: Chagee already plans this part.
  YF Chuhai; Chagee Form 20-F
- **Volatile downstream**
  Demand at a counter moves by the hour: lunch, a mall event, rain, a new drink launch.
- **So we stop at the counter**
  The open question is not the shipment. It is how many people the counter needs this hour, and how they work.

TAKEAWAY | Chagee already plans its stock. The open problem is matching people to the hourly peak.

SPEAKER NOTES · FROM FARM TO CUP

Press play and the cup travels farm to customer. Sourcing is standard; only fresh milk and fruit are local (YF Chuhai). Chagee runs central and regional warehouses (Form 20-F) and already plans restocking from forecasts.

The volatile part is the counter, the red chips. Under time pressure, summarise this slide in one sentence: supply is planned, the counter is not. About forty seconds.

Every cup is already a timestamped record: the inputs for a forecast exist today

Four data sources and six decisions they could feed. We follow the four that start in the app and end at the counter.

WHICH SOURCE FEEDS WHICH DECISION · STEP THROUGH

	Store and production	App and members	Delivery	Supply and context
Demand forecasts	Cups per hour by drink and channel	Member repeat demand	Orders per hour by platform	Weather, holidays, launches
Staffing and prep	Wait times, machine logs	Pickup no-shows	Hand-over peaks	Labour hours vs cups per hour
Replenishment	Drink mix per outlet	Launch trial rates	Group-order toppings	Ingredient use, expiry, par
Queue and capacity	Wait time, footfall vs sales	Store-select drop-offs	Rider wait at counter	Stock-outs at peak
Site selection	Sales density per outlet	"No store nearby", home and work	Distance and fee by postcode	Mall footfall, competitors
Menu and promos	Drink share, customisation	Trial and repeat, voucher uplift	Delivery basket mix	Collab calendar

White chips mark the core input; grey text refines it. Red rows: our scope.

- **No new sensors needed**
  Point-of-sale, app and machine logs already exist. The work is joining them by outlet and hour.
- **The app is the link**
  Only the app ties a cup to a person. That needs consent and care under Singapore's data protection law (PDPA).
- **What we take forward**
  Demand forecast, then staffing, then queue capacity: one chain that ends with the GM's counter and roster.

TAKEAWAY | The data for an hourly forecast is already collected. What is missing is turning it into a roster.

SPEAKER NOTES · THE DATA TRAIL

Each step lights up one decision and the data that feeds it. Skip to the last step under time pressure: our scope is the three counter rows: forecast, staffing, queue.

This list is our reading of what Chagee's systems record, based on public material and our industry conversation, not a Chagee data dictionary. About fifty seconds.

New stores and the in-store peak, for one decision-maker: the GM

Two of our three ideation areas, enabled by two kinds of AI, each simulated with the trade-off the GM weighs.

FROM THREE IDEAS TO TWO · STEP THROUGH

NOT COVERED YET

Online retailing

Steer orders between app pickup and Grab by hour.

To be added. Needs platform fees and price tests we do not have yet.

OUR FOCUS · EXPANSION

New stores

Map where the next outlet adds the most cups, from app searches.

Rent, fit-out and payback stay future work.

Topic 3 · forecasting

OUR FOCUS · OPERATIONS

In-store operations

Forecast the peak, set the roster, share the counter.

Data exists, the decision is daily.

Topic 3 · forecasting

Topic 5 · bucket brigades

AI · PREDICTIVE

Forecast demand: where a new outlet adds cups, and cups per hour for the roster.

AI · PHYSICAL

Automation and machines at the counter, from dispensers up to machines that make the drink.

OPERATIONS TOOLS, NOT AI

Bucket brigade and cross-training: how people share the hand steps.

The GM

General manager, Chagee Singapore

DECIDES	Where to open next, which counter steps go to machines, how many people work the peak, and how they share it
WHEN	Before a store opens, and when rosters are set
USES	The forecasts and the simulations
TODAY	To confirm with Chagee: how rosters are set now

SIMULATED, WITH THE TRADE-OFF THE GM WEIGHS

PREDICT	cups added vs cups moved from nearby outlets
SHARE	bucket brigade vs fixed posts
SPLIT · MACHINES	shorter wait vs higher cost
BENEFIT	orders kept vs extra labour

TAKEAWAY | Predictive AI says where to open and how many people the peak needs; physical AI changes which steps machines do; the bucket brigade decides who does the rest.

SPEAKER NOTES · OUR SCOPE

Our ideation produced three areas. Online retailing is not covered yet and will be added; for now we take two: new stores, for expansion, and the in-store peak.

AI comes in two kinds here. Predictive AI forecasts demand: where a new outlet adds cups, and how busy each hour will be. Physical AI is automation and machines at the counter, from dispensers up to machines that make the drink; that is a type of machine, not a claim about Chagee's plans. Bucket brigade and cross-training are operations tools from Topic 5, order fulfilment, but they decide whether a lean roster actually works.

Every part of the deck is a simulation with a trade-off the GM has to weigh: cups added against cups moved, brigade against fixed posts, a shorter wait against a higher cost, and orders kept against extra labour. The decision-maker is the GM; we do not yet know how Chagee sets rosters today. About fifty seconds.

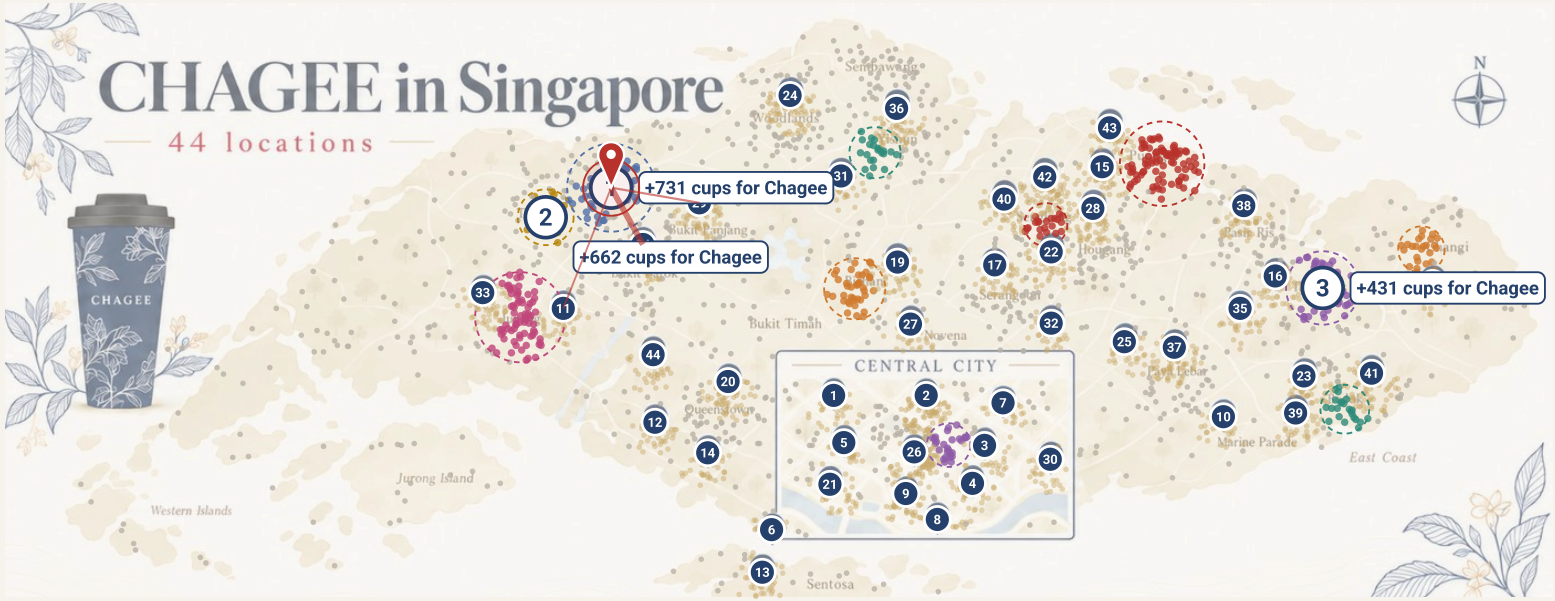
Course link · Week 3, Analytics and AI · the Flashion new-store forecast

Where would outlet 45 add the most cups for Chagee?

Not what a new store sells, but what it adds once you subtract the cups it takes from nearby outlets.

CONTEXT Chagee is looking to expand rapidly in Singapore; how many stores it is going for is confidential.

NEW-STORE FORECAST



SUGGESTED SITE 1 · 2.7 KM FROM 34 · WEST MALL

Strong site: mostly new cups

+731 cups a day for Chagee

The new store sells 837 a day (1.5× an average outlet). 106 of those move from outlets nearby:

34 · West Mall, 2.7 km	**−77**
29 · Bukit Panjang Plaza, 3.3 km	**−28**

Peak hour 13:00: **92 cups** → **2 people** on today's counter (slides 06–07).

Synthetic app data on the team's map; outlet positions approximate (the central inset is enlarged). Not a trained model: the point is the method.

WITHOUT AI our assumed baseline

Open outlet 45 beside 34 · West Mall: it sells **604** cups a day, but 64% are moved from outlets nearby, so **+216** for Chagee.

WITH AI our proposal

Predictive AI on app searches (DBSCAN) puts the same outlet 2.7 km away, near Choa Chu Kang: it sells 837, **+731** for Chagee.

SPEAKER NOTES · PREDICT DEMAND FOR A NEW STORE

Chagee Singapore runs on its app, with few kiosks, so every search is a demand signal, including searches where no store is close. That makes the app the forecasting input for a new store, before it has any sales.

Step one: synthetic app sessions on our map. Gold are within walking distance of an outlet; red found no store nearby. Step two: DBSCAN, unsupervised machine learning, groups red searches that are dense; scattered ones are noise. Step three: cluster centres are candidate sites.

Step four: click the map to test a site. The card leads with the gain for Chagee, which is the store's sales minus the cups it takes from nearby outlets, and gives a verdict: strong, mixed, or mostly moving cups. Try a pin next to Tampines West CC: it sells, but most of it moves from the two Tampines outlets nearby. Step five: the forecast's peak hour gives the opening roster for the counter on the next slides.

If asked how DBSCAN works, press How it works: it replays the algorithm one search at a time on a toy set. The appendix lists every simulated number.

The three sliders are the only inputs: cluster radius, minimum searches per cluster, and how far people will walk. The data is synthetic; with Chagee's real app logs, the same steps would run unchanged. About one and a half minutes.

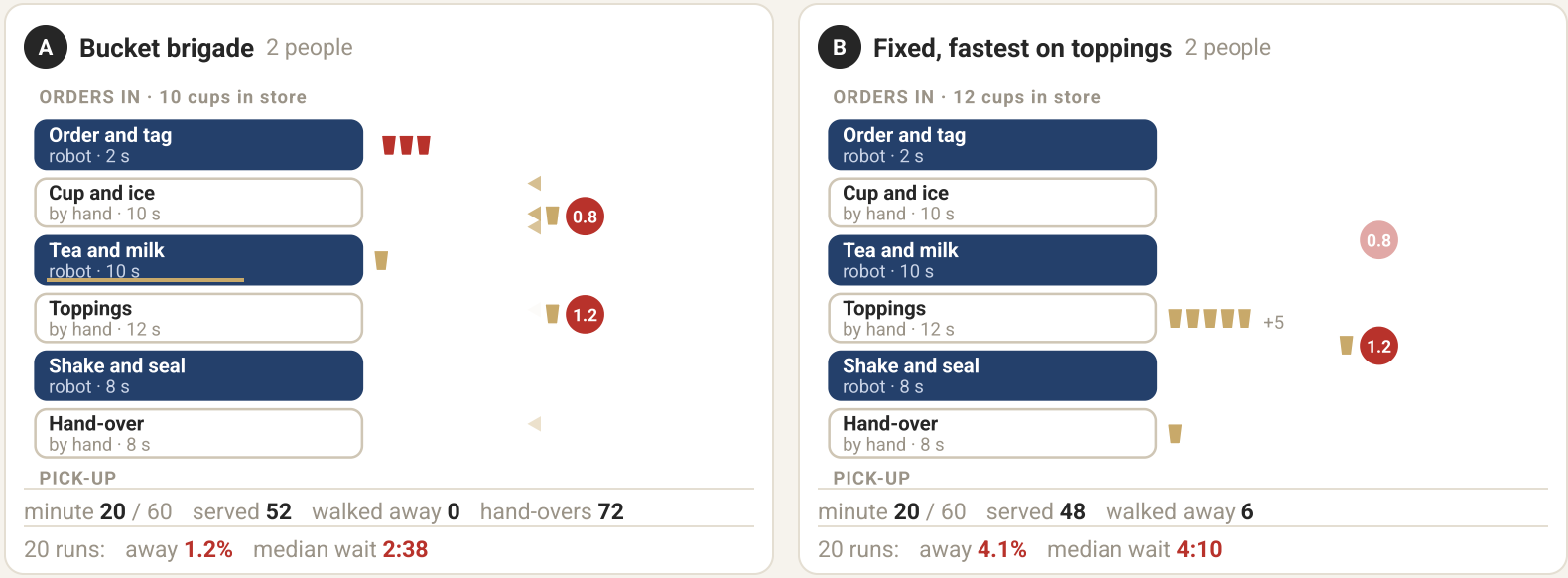
Course link, Week 3, Analytics and AI: in the Flashion example, demand for a new store was forecast from similar stores. We ask the same question with a different input: app searches around the site.

Course link · Week 5 and Week 6, equipment and technology · the bucket-brigade simulation

With a lean roster, a bucket brigade keeps cups moving without knowing who is fastest

Today's counter, two ways of sharing the hand steps, side by side. Pick what each side runs.

PEAK HOUR · SAME CUSTOMERS ON BOTH SIDES



Red circle = a person (number = speed) · gold ring = held back · faded = walking · ▲ = hand-over · navy = robot · red cup = waiting for a person

ALL THREE RULES, MEAN OF 20 RUNS

EVENT DAY, 2 PEOPLE	WALKED AWAY	MEDIAN WAIT
Bucket brigade	**1.2%**	**2:38**
Fixed, fastest on toppings	4.1%	4:10
Fixed, slowest on toppings	34.9%	9:47

At the people on side A. A customer leaves if the wait would pass 5 min. With spare people, a well-assigned fixed line is as good or a little faster.

- **Why lean is where it counts**
  Work flows to whoever is free, so the line balances itself (Topic 5). Fixed stations are only as good as the guess of who is fast.

WITHOUT AI our assumed baseline
A normal-day roster (1 at the peak) on an event day: **54%** of peak-hour orders walk away.

WITH AI our proposal
The forecast sets the crew (2) and a bucket brigade shares it: **1.2%** walk away, without knowing who is fastest.

SPEAKER NOTES · SHARE THE COUNTER

Same counter, same customers, two sharing rules side by side. Robots, the navy boxes, take the order, brew tea and milk, and shake and seal. People do cup and ice, toppings and hand-over.

Left, a bucket brigade: each person carries a cup forward; the last hands it over, walks back and takes over a colleague's cup. Right, fixed stations: each person owns a station.

Press play on an event day. Then set the right side to "slowest on toppings": the queue builds at toppings and customers walk away. The brigade never needed to know who was fast.

Be honest about the other case: add people and fixed stations catch up, or are a little faster. About one minute.

Course link: the bucket-brigade simulation from Week 5 and Week 6, equipment and technology, inspired this counter replica. Same idea, a different use case.

Takeaway: At the forecast's roster on an event day, the brigade loses about 1.2% of orders; fixed stations lose 4.1% to 35%, depending on who is on toppings.

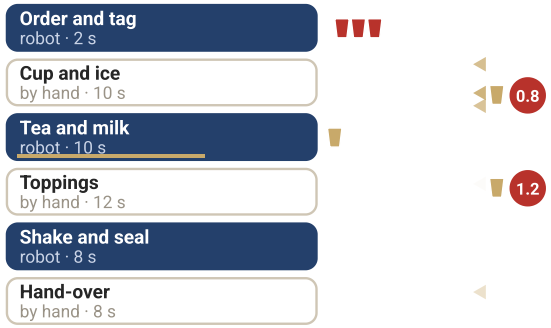
Course link · Week 5 and Week 6, equipment and technology

Keep today's split: machines on the fixed steps, people on the rest. Robots buy a shorter wait, not a lower cost

PEAK HOUR · SAME CUSTOMERS

A Today's counter 2 people, brigade

ORDERS IN · 10 cups in store

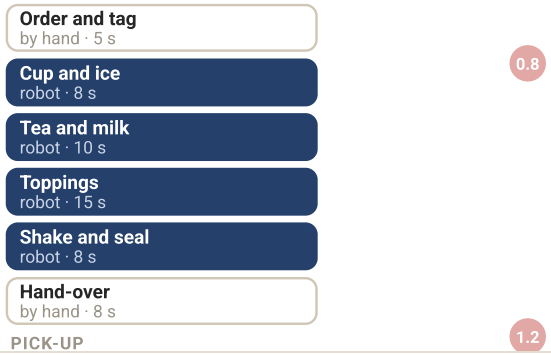


minute **20** / 60 served **52** walked away **0** hand-overs **72**

20 runs: away **1.2%** wait **2:38** **S$402**/day

B Machine-made drink (scenario) 2 people, fixed posts

ORDERS IN · 0 cups in store



minute **20** / 60 served **57** walked away **0**

20 runs: away **0.0%** wait **1:06** **S$465**/day

Navy = robot, white = by hand (click to switch) · red circle = a person · faded = walking · red cup = waiting for a person.

ALL FOUR SPLITS, MEAN OF 20 RUNS

EVENT DAY	PEOPLE	WALKED AWAY	WAIT	COST/DAY
Today's counter ✓	**2**	**1.2%**	**2:38**	**S$402**
Machine-made drink (scenario)	2 fixed	0.0%	1:06	S$465
All by hand	4	0.0%	1:25	S$422
All robots	–	0.0%	0:58	S$562

✓ cheapest within the limit · fastest: All robots

Fewest people and the better sharing rule keeping walk-aways within 2%; cost = people hours × wage + robots.

Normal day, cost a day: All by hand S$361 (cheapest) · Today's counter S$372 · Machine-made drink (scenario) S$435 · All robots S$562. Fastest: All robots (0:52); today's counter is slower than all by hand (2:09 vs 1:16). **Which machine?** Four splits only: one tea-and-milk machine pays at 556 cups a day (+S$1,706/month); robots wait for about 757 cups (machine ladder, appendix).

WITHOUT AI our assumed baseline — All by hand on an event day: **4 people**, median wait 1:24, **S$422** a day.

WITH AI our proposal — Machines on the fixed steps (Today's counter): **2 people**, **S$402** a day, but a longer wait (2:38); all robots wait 0:58 for S$562.

GM DECISION Keep machines on order and tag, tea and milk, shake and seal: the cheapest split within the walk-away limit on an event day (S$402), and only S$10 a day above all by hand on a normal day. Do not buy the machine-made drink yet: S$64 a day more for a 1:32 shorter wait. Revisit at about 757 cups a day.

SPEAKER NOTES · SPLIT THE COUNTER

This is the GM's decision: which steps of the counter stay human and which go to robots. Side B, machine-made drink, is our scenario, not a claim about Chagee: machines make the drink while people tag the order and hand it over. Chagee's own equipment cost is not public; its Singapore relaunch used semi-automated equipment, according to Vulcan Post.

Click any station to switch it. Each side gets its forecast roster; change people with plus and minus. The table on the right is the simulation run for all four splits.

What it shows, honestly: the table is the event day; the line under it is the normal day, at about 556 cups. On a normal day all by hand is the cheapest of these four splits and all robots the fastest, so robots buy a shorter wait, not a lower cost; today's split is in fact slower than all by hand that day. On an event day, today's split is cheapest within the limit. The machine-made drink scenario needs a second person at the peak, because one person cannot tag and hand over at opposite ends of the counter.

These are four splits only. A single machine can undercut all by hand, because it lets the roster drop a person off-peak; the machine ladder in the appendix has the prices and payback. About one and a half minutes.

Course link: the same bucket-brigade replica from Week 5 and Week 6, with machines added on some stations.

Takeaway: On an event day, Today's counter is the cheapest split within the walk-away limit, at about S$402 a day. Machine-made drink (scenario) needs 2 people at fixed posts and costs about S$64 more, for a median wait of 1:06.

How to use it: choose a split for each side, or click any station to hand it to a robot or a person, then run the peak hour or press End to compare.

Choose the split, roster to the forecast, and share the work to suit the split

Three moves for the GM, each with a trigger, then a pilot to test them.

WHAT THE FORECAST CHANGES, PEAK HOUR, TODAY'S COUNTER

DAY	FORECAST PEAK, CUPS PER HOUR	PEOPLE AT THE PEAK	WALK-AWAYS, FORECAST ROSTER AND BRIGADE	WALK-AWAYS, NORMAL-DAY ROSTER
Normal day	61	**1**	0.9%	0.9%
Busy day	92	**2**	0.0%	**20.8%**
Event day	153	**2**	1.2%	**53.8%**

1 Choose the split

Robots where speed matters more than cost; people where the work varies: tagging and hand-over.

2 Roster to the forecast

On today's counter, one more person for about every 83 forecast cups in the peak hour.

3 Share to suit the split

A brigade when hand steps sit along the line; fixed posts when machines make the drink.

- **Next step: pilot in 3 outlets**

Shadow	Roster	Roll out
forecast only	staff to it	if KPIs hold

KPIs: cups per labour hour, wait time, walk-aways at the peak. Compare forecast and actual cups weekly.

- **Limits we know**

All figures are illustrative assumptions, not Chagee data. Robot speeds and costs are assumed; breakdowns and breaks are not modelled.

- **To confirm with Chagee**

How rosters are set today, robot costs per station, and whether outlets log wait times and walk-aways.

WITHOUT AI our assumed baseline — One roster for every day: **21%** of peak-hour orders walk away on a busy day, **54%** on an event day.

WITH AI our proposal — Roster to the forecast (2 on busy, 2 on event days): **0.0%** and **1.2%** walk away.

SPEAKER NOTES · THE GM'S PLAYBOOK

Three moves for the GM: choose the split, roster to the forecast, and share the work to suit the split. Then a pilot in 3 outlets: shadow, roster, roll out, judged on cups per labour hour, wait time and walk-aways.

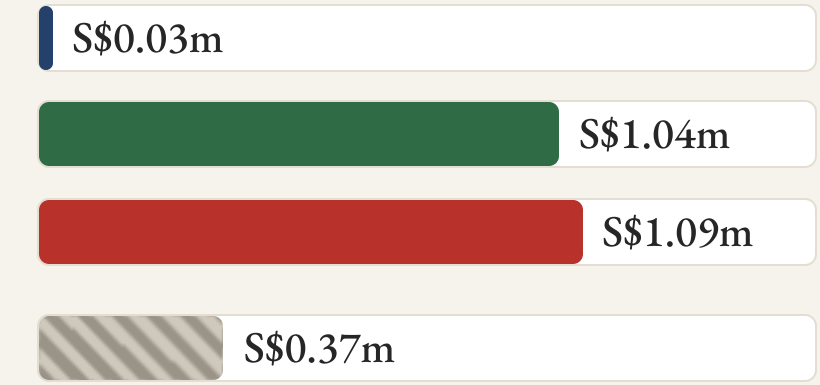
Close on the limits: these are our assumptions, not Chagee data. New-store openings, the next part of our work, will use the same simulator before a store opens. About one minute, then questions.

Takeaway: Predictive AI sets how many people; the split sets what robots do; the sharing rule sets who does what. Next, test it in 3 outlets.

Across Singapore's 44 outlets, about S$1.09m a year: the machine saves labour, the forecast keeps orders

Modelled avoided costs on stated assumptions, not demonstrated savings. Against a flat normal-day roster and an all-by-hand counter; we do not know how Chagee rosters today.

PROFIT A YEAR, ALL 44 OUTLETS



Profit = orders kept × price × margin, plus labour saved, minus machine cost. Calendar: 104 busy days (weekends) and 11 event days (public holidays) a year, an illustrative definition. Peak hour only, so orders kept are a floor. Every figure, range and rule: appendix.

PREDICTIVE AI · ROSTER TO THE FORECAST

Keeps orders, but staff cost most of the margin

About **54,668** peak-hour orders a year kept across Singapore, mostly on event days (61 per outlet that day). That is S$0.32m of revenue but only S$0.03m of profit, or S$0.08m if a person is added only when it pays.

PHYSICAL AI · ONE TEA-AND-MILK MACHINE

Saves labour: the big number

Lets the roster drop from 2 people to 1 in the off-peak hours: **S$1.04m** a year after the machine's cost.

READ WITH CARE

The big number rests on one assumption

The minimum crew, 2 by hand and 1 with machines, is ours. Without it, S$1.09m falls to **S$0.37m**.

WITHOUT AI our assumed baseline	Flat normal-day roster, all by hand: about **1,291** peak-hour orders a year walk away per outlet.
WITH AI our proposal	Both levers, one run: **1,278** kept a year per outlet, profit **S$1.09m** across 44 outlets (S$0.37m without the minimum-crew rule).

SPEAKER NOTES · THE SIZE OF THE BENEFIT

One number and where it comes from. Across Singapore's 44 outlets the model gives about S$1.09m a year, against a flat normal-day roster and an all-by-hand counter. These are modelled avoided costs on our assumptions, not demonstrated savings.

Read the bars top to bottom. The forecast on its own keeps orders, about 54,668 a year, mostly on event days, but the extra staff it rosters cost most of the margin. The machine on its own is the big number, because it lets the roster drop a person off-peak. Both together is the headline.

Say the caveat out loud: the machine's saving comes from our minimum-crew assumption. Without it, the total falls to about S$0.37m. The detail, ranges and both crew rules are in the appendix. About one minute.

THANK YOU

QUESTIONS AND DISCUSSION

OPIM639 · GROUP 3 · KEZIAH, JIN, AYUMI, REVAN



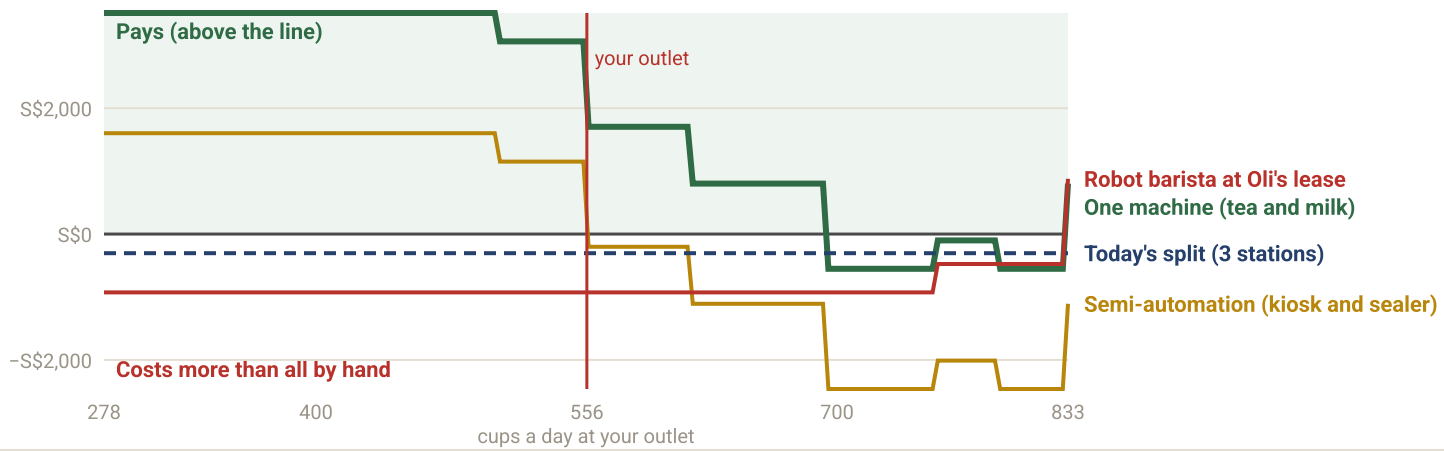
SPEAKER NOTES · THANK YOU

Likely questions: is the brigade always better? No; with spare staff fixed stations are as good. Is the forecast real? No, it is an illustration of the chain, built on stated assumptions.

A machine pays only when it removes a person-shift for less than the shift costs

To our knowledge, based on public sources; vendor list prices, not cost to implement. Chagee's equipment cost is not public; its Singapore relaunch used semi-automated equipment (Vulcan Post).

NET GAIN A MONTH AGAINST ALL BY HAND · ONE OUTLET



Shown at 556 cups a day, wage S$15.06 an hour, S$1,909 a month per machine unit (sliders on screen).

	1 Semi-automation	2 Beverage platform	3 Milk-tea machine	4 Robot arm	5 Robot barista
	Gong Cha Singapore	Gong cha Super Wu	Amacha, one shop	Richtech ADAM	Oli; coffee-first
	Not disclosed	*Not disclosed*	S$10,000 to buy	*Not disclosed, rental only*	US$4,990/mo lease
	PRESS	COMPANY CLAIM	PRESS	VENDOR PAGE	VENDOR PAGE

The model's S$1,909 a month is per station unit (our reading: a semi-automated station). Semi-automation and the beverage platform are priced at it because their real prices are not disclosed; robot prices are per setup. Full table and sources in the appendix.

START HERE

One machine on the tea and milk

+S$1,706 a month vs all by hand

Saves 8 person-hours a day (S$3,614 a month) for one S$1,909 unit. Simple payback, if it lets the roster drop one person-shift: 2.8 months on Amacha's S$10,000.

WATCH

Today's 3-station split

−S$305 a month vs all by hand

Saves 12 person-hours a day (S$5,422) but pays for 3 units (S$5,726).

WAIT FOR VOLUME

Robots: arm or barista

757+ cups a day before they beat today's split on labour

At 556 cups, Oli's lease (S$6,350) is S$928 a month worse; a robot setup would need to cost under S$5,422 a month. The arm's price is not disclosed.

Model assumption that drives all three: minimum crew 2 by hand, 1 with machines, off-peak. Station speeds are the model's, not vendor claims. Rosters step in whole people, so results flip with volume and wage.

WITHOUT AI — our assumed baseline

All by hand at 556 cups a day: **24 person-hours** a day, **S$10,843** a month of labour.

WITH AI — our proposal

One tea-and-milk machine: **16 person-hours** a day, **+S$1,706** a month after its unit cost. Robots wait for about 757 cups a day.

SPEAKER NOTES · MACHINE LADDER

Backup for slide 07, for questions on which machine to buy. This is physical AI: automation and machines at the counter. The predictive AI is the forecast that sets the roster. One rule explains the whole slide: a machine pays only when it removes a person-shift, and costs less than that shift.

The chart shows the net gain a month against an all-by-hand counter as volume changes; above the line it pays. One tea-and-milk machine lets the roster drop from 2 people to 1 in 8 off-peak hours, about one person-shift a day, for one station unit: it pays across most of the range. Today's split saves 12 person-hours but pays for 3 units, so it ends up a little dearer than all by hand. A robot saves no more labour than today's split until about 757 cups a day, so at Oli's lease it only pays at the top of our range.

Three things to say out loud. The minimum crew, 2 by hand and 1 with machines, is our model's assumption and drives this result. Every option runs at the model's station speeds, not vendor claims. Payback is simple payback, if the machine lets the roster drop one person-shift, with no installation, maintenance or financing. Chagee's own equipment cost is not public. About one minute.

Takeaway: At 556 cups a day: start with one machine (about +S$1,706 a month); today's 3-station split costs S$305 a month more than all by hand; robots save no more labour than today's split until about 757 cups a day.

Per outlet, about 61 orders kept on an event day and 5 on a busy day; on an illustrative calendar, S$7,539 of revenue a year

Modelled avoided costs on stated assumptions, not demonstrated savings. Peak hour only; conservative. Prices to our knowledge, based on public sources.

Profit a year, 44 outlets: **S$1.09m** = counter automation, physical AI **S$1.04m** (depends on the minimum-crew rule) + forecast roster, predictive AI **S$0.05m** | Without / with the minimum-crew rule: **S$0.37m to S$1.09m**

ONE OUTLET AT 556 CUPS A DAY · BOTH LEVERS, ONE RUN

EVENT DAY
61 orders kept
Revenue S$358 (× S$5.90) · profit S$241
Counter: labour saved − machine S$27

BUSY DAY
5.4 orders kept
Revenue S$32 (× S$5.90) · profit S$76
Counter: labour saved − machine S$57

NORMAL DAY
0.2 orders kept
Revenue S$1 (× S$5.90) · profit S$58
Counter: labour saved − machine S$57

A YEAR	PER OUTLET	44 OUTLETS, IF ALL LOOKED LIKE THIS MODEL
Orders kept peak hour, busy and event days	**1,278** range −121 to 1,501	**56,221**
Revenue orders × price	**S$7,539** range −S$832 to S$10,357	**S$331,704**
Profit margin 50%–70%: −S$754 to +S$754	S$24,660	S$1,085,027
Labour saving, counter labour saved − machine cost; shown on its own	S$20,136	S$886,005

Revenue and labour are never added: revenue is orders × price; profit is orders × price × margin, plus labour saved, minus machine cost.

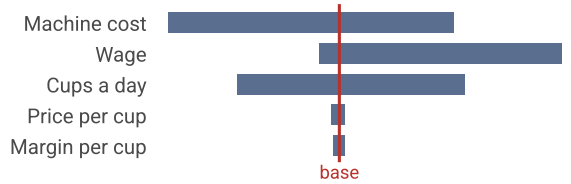
Calendar: 104 busy days (weekend days) and 11 event days (Singapore public holidays) a year: an illustrative definition, to confirm with Chagee; sliders on screen.

ORDERS · PROFIT A YEAR	WALK-AWAY RULE	ECONOMIC RULE
Forecast crew	1,242 · S$603	670 · S$1,711
Machine	888 · S$23,611	888 · S$23,611
Both, one run	**1,278 · S$24,660**	**1,278 · S$24,660**
Both, today's split counter −S$305/month	1,124 · −S$3,142	1,124 · −S$3,142

Walk-away rule: fewest people keeping peak-hour walk-aways within 2% (the forecast crew at 556 cups a day). Economic rule: add a person only when orders kept × price × margin beat the extra labour. "Both" is one run, not a sum: the levers keep many of the same orders, and with the machine the forecast crew needs fewer extra hours (+S$446 against the sum).

Against what
A flat normal-day roster and an all-by-hand counter. We do not know how Chagee sets rosters today (to confirm with Chagee).

All inputs at their worst/best end together
Profit a year, one outlet: low **−S$15,704** · base **S$24,660** · high **S$101,166**



Fragile: the minimum crew
The counter lever rests on the model's minimum crew off-peak: 2 by hand, 1 with machines. Profit a year per outlet: S$24,660 with it; S$8,395 with 1 either way.

Not sized
Delivery levers (third-party; channel margins not disclosed), membership and CRM, waste (no inventory model in this deck), and the new-store site forecast (shown as cups added only).

WITHOUT AI our assumed baseline — Flat normal-day roster, all by hand: about **1,291** peak-hour orders a year walk away per outlet.

WITH AI our proposal — Both levers, one run: **1,278** kept a year per outlet, profit **S$1.09m** across 44 outlets (S$0.37m without the minimum-crew rule).

SPEAKER NOTES · THE BENEFIT IN DETAIL

Backup for slide 09. This answers part c of the brief: how big is the benefit, for Singapore only. Against what: a flat normal-day roster and an all-by-hand counter. We do not know how Chagee rosters today, so that is a question for Chagee.

Lead with orders. On an event day the model keeps about 61 orders per outlet in the peak hour that would otherwise walk away, about S$358 of revenue; on a busy day about 5. Over a year it depends on how many such days there are: our illustrative calendar is 104 busy days, the weekends, and 11 event days, the public holidays. That is an input, not a measurement.

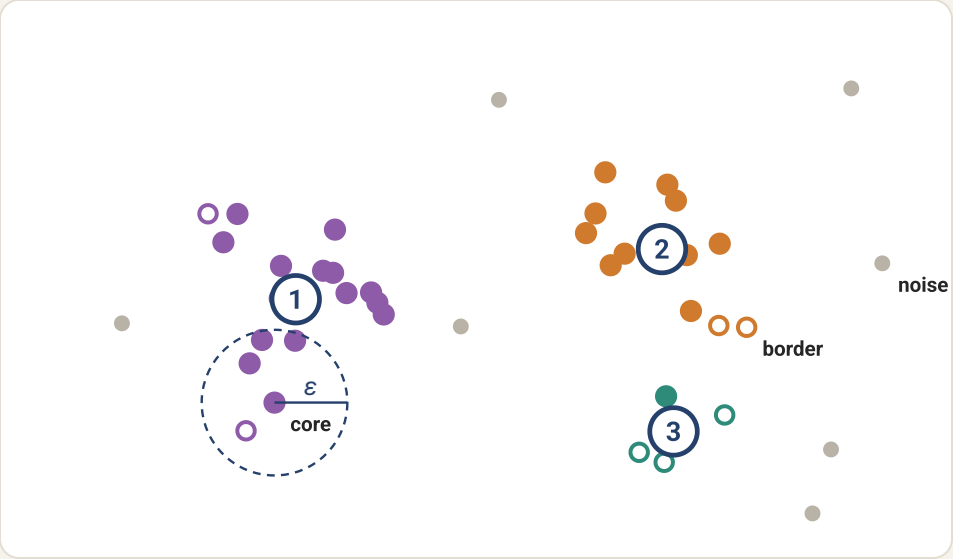
Profit is the second line, using a margin we assume because Chagee does not disclose one. The labour saving is shown separately and never added to revenue. The two levers are run together, not added, because they share the same labour hours. The strip at the top splits the profit: most is counter automation, physical AI, the machine on its own; the forecast, predictive AI, adds a small share on top.

Two crew rules for rostering to the forecast. The walk-away rule adds people until peak-hour walk-aways are within 2%; at our default volume that is exactly the forecast crew. The economic rule adds a person only when the orders kept are worth more than the extra hours. They differ on busy days by hand, where the economic rule adds no one.

Be honest about fragility: most of the profit is the counter's labour saving, and it comes from the model's minimum crew, 2 by hand and 1 with machines. Drop that rule and the profit falls to about S$8,395 a year per outlet. And rostering to the forecast pays on event days, not on busy days, where the extra hours cost more than the orders won back. These are modelled avoided costs on stated assumptions, not demonstrated savings. About one minute.

DBSCAN finds where searches bunch up; a proximity model turns a pin into cups

DBSCAN IN FOUR MOVES · TOY EXAMPLE



1. **Radius ε.** Two searches are neighbours if they are within ε of each other (the cluster-radius slider).
2. **Core.** A search with at least the minimum number of searches within ε (counting itself) is a core point.
3. **Grow.** A cluster spreads from core to core. Searches it reaches that are not core join as border points.
4. **Noise.** Searches no cluster reaches. Unlike k-means, nobody picks the number of clusters, and odd shapes are fine.

Unsupervised: no labels or past openings are needed. Ester, Kriegel, Sander and Xu (1996). On slide 05 it runs only on searches with no outlet within walking distance; each cluster's centre is a candidate site.

THE SIMULATED NUMBERS, IN ORDER · AT THE SLIDER DEFAULTS

1	**App searches**	Synthetic dots on land: 15% spread evenly, 35% around the outlets' malls, the rest around 20 towns	**3,000 dots**
2	**No store nearby**	The nearest outlet is further than the walking distance	**1 km · 46% of dots**
3	**DBSCAN**	Clusters the no-store-nearby dots	**ε 0.6 km · min 15 · 11 clusters**
4	**Pull of a store**	Fades with distance d: exp(−½ (d/L)²), with the length L set by the walking distance	**L = 1 km**
5	**Who buys where**	A dot buys at a store in proportion to its pull, against the pull of not buying (a store next door = one)	**not buying 0.3**
6	**Calibration**	Each dot's weight is set so the average outlet sells the deck's cups a day (cups a year ÷ open days)	**556 cups a day**
7	**Score a pin**	Cups for Chagee = what the new store sells − what it moves from nearby outlets	**top: 837 − 106 = +731**
8	**Verdict**	Share of the new store's cups that are new to Chagee	**strong ≥ 60% · weak < 40%**
9	**Opening roster**	Peak-hour cups ÷ cups one person handles, with the safety margin	**one per 83 cups**

All synthetic: the dots, towns and weights are assumptions on the team's map (not to scale), not Chagee app logs or population data. With real app logs, steps one and two change; the rest runs unchanged.

SPEAKER NOTES · HOW THE SITE FORECAST WORKS

Backup for questions on slide 05. Left: DBSCAN in four moves on a toy set. A search is a core point if enough searches sit within the radius; clusters grow from core to core; border points join; the rest is noise. We do not choose the number of clusters.

Right: every simulated number, in the order the model uses it. The forecast is only as good as these assumptions; with Chagee's real app logs, the first two rows would be replaced by data and the rest would run as is.

Every rung, every price as published, and where it comes from

To our knowledge, based on public sources. Physical AI: automation and machines. A gap is "not disclosed", never a guess.

RUNG	EXAMPLE	AUTOMATES · STEPS ON MACHINES (OUR MAPPING)	VENDOR LIST PRICE	CLAIMS (NOT MODEL INPUTS)	RELIABILITY	SOURCE
1 **Semi-automation**	Gong Cha Singapore	Sugar dispensers, cup sealers, self-order kiosks. **Order and tag, Shake and seal**. Kiosk = order and tag, sealer = shake and seal. Sugar dispensing sits inside tea and milk, so it is not modelled.	*Not disclosed*	Over 20% productivity from self-ordering; 5% manpower saving (no baseline given)	PRESS	https://www.ntuc.org.sg/uportal/news/Technology-Brewing-Up-the-Bubble-Tea-Business/
2 **Beverage platform**	Gong cha Super Wu (Feb 2026)	Dispenses tea, sweeteners and syrups; brewing and shaking stay manual. **Tea and milk**	*Not disclosed*	Up to 65% higher peak productivity	COMPANY CLAIM	https://marketech-apac.com/gong-cha-launches-gong-cha-2-0-to-modernise-global-operations-and-store-design/
3 **Single automated milk-tea machine**	Amacha (one small shop, owner interview)	Makes the milk tea itself. **Tea and milk**	S$10,000 to buy	The shop's total opening cost was S$209,000 (not a machine cost)	PRESS	https://www.malaymail.com/news/life/2026/10/05/opening-a-milk-tea-shop-in-singapore-can-rack-up-to-s209000-even-before-the-first-cup-is-poured/237715
–	**Our scenario: today's split**	**Order and tag, Tea and milk, Shake and seal**. A scenario of ours (slides 06 and 07), not Chagee's disclosed setup	Model: S$1,909 a month per station unit		OUR ASSUMPTION	assumptions.json
4 **Robot arm that makes boba**	Richtech ADAM	Cup, ice, tea, toppings and sealing by one robot arm. **Cup and ice, Tea and milk, Toppings, Shake and seal**	*Not disclosed*. Rental only; price not published	–	VENDOR PAGE	https://richtechrobotics.com/solutions/adam
5 **Robot barista**	Oli	A whole drink station in one unit. **Cup and ice, Tea and milk, Toppings, Shake and seal**. Coffee-first. No tea drinks listed for Oli or Artly.	US$59,000 (≈ S$75,078) to buy; US$4,990 (≈ S$6,350) a month lease	–	VENDOR PAGE	https://olirobotics.com/best-robot-barista
	Artly		US$68,000–88,000 (≈ S$86,530–111,980) to buy; US$2,500–4,000 (≈ S$3,181–5,090) a month lease; US$2,500–4,000 a month software Oli's estimate of a competitor. The figures as given do not say whether the lease replaces or adds to the purchase, so no verdict is computed.	–	VENDOR PAGE	https://olirobotics.com/best-robot-barista
	Cafe X		US$250,000 (≈ S$318,125) to buy Oli's estimate of a competitor's price.	–	VENDOR PAGE	https://olirobotics.com/best-robot-barista
–	**Chagee Singapore**	Chagee's own equipment cost is not public. Its Singapore relaunch used semi-automated equipment.	*Not public*	–	PRESS	https://vulcanpost.com/872176/chagee-singapore-comeback-after-sudden-exit/

Vendor list prices, not cost to implement: installation, training, maintenance, power, space and import are not included; USD converts at S$1.2725 per US$. **Simple payback** = list price ÷ labour saved a month, with no maintenance or financing. **The model's S$1,909 a month** is per station unit: our reading: one semi-automated station unit (for example a dispenser or a cup sealer), amortised with upkeep. Not a robot barista lease; robot prices in machine_tiers.json are per setup. **Vendor speed and throughput claims** are listed, never used: every rung runs at the model's station speeds. **Steps on machines** is our mapping. Artly and Cafe X prices are Oli's estimates of competitors.

SPEAKER NOTES · MACHINE LADDER: TIERS AND SOURCES

Backup for the machine ladder. Every price is as the source publishes it; where a source gives no price we say not disclosed. Claims such as Gong cha's productivity figures are listed for context and are not inputs to our model. Soft copy only.

Where every number comes from

FACTS QUOTED

Outlets in Singapore: 44. Store count used on the storyboard deck. The doc also cites 46 (VnExpress, earlier); the later directory count is used. *[L-sg-directory]*

Franchise share of FY2025 net revenue: 0.88. Secondary source; flagged wherever shown. *[L-koalagains]*

Company-owned share of FY2025 net revenue: 0.12. Secondary source; flagged wherever shown. *[L-koalagains]*

SOURCES

L-sg-directory Chagee Singapore store directory, mapped by the team (44 locations, checked 15 Oct 2026); Mothership, Sep 2026, https://mothership.sg/2026/09/chagee-more-heartland-stores-singapore-eugene-lee/ gives the same count. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

L-koalagains KoalaGains, 'CHA business and moat', https://koalagains.com/stocks/NASDAQ/CHA/business-and-moat (SECONDARY - flag when used): FY2025 net revenue about 88% from the franchise (master franchise) model and about 12% from company-owned stores, the latter growing fastest. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

L-yfchuhai YF Chuhai, https://www.yfchuhai.com/article/9751.html: Chagee's leaf supply chain is in China; it sources only two ingredients locally, fresh milk and fruit. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

L-sec20f Chagee Holdings Form 20-F for FY2025, SEC EDGAR, https://www.sec.gov/Archives/edgar/data/2013649/000110465926050766/cha-20251231x20f.htm: separate centralized and regional warehouses to lower storage and logistics costs. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

L-unsw UNSW BusinessThink, Oct 2024: Chagee semi-automation, baristas scan a QR code and the machine dispenses the formulation. (brief.md section 9)

I-chagee-contact INFORMAL - a Chagee contact (Oct 2026). Three points only: whether Singapore outlets are owned, a joint venture or franchised does not matter for this work; Singapore outlets run under one company and one data system; all delivery is via third-party apps. Confirm with Chagee before citing.

I-sg-groundfloor INFORMAL - quote in the team's ideation doc (Chagee Singapore's Lee: 'Whenever you see a Chagee, it's always on the first floor'; kiosk-style outlets in basements and MRT stations at entry in 2019). The article is not named in the doc; confirm the source before quoting.

I-team-app-first INFORMAL - team observation in the ideation doc: Chagee Singapore sells mainly through its app, with minimal order kiosks in outlets. Confirm before citing.

L-ester1996 Ester, M., Kriegel, H.-P., Sander, J. and Xu, X. (1996). A density-based algorithm for discovering clusters in large spatial databases with noise. Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD-96), 226-231. The DBSCAN algorithm used on slide 05.

P-honeycombers PRESS - in-store sightings, indicative. Honeycombers (Feb 2026), https://thehoneycombers.com/singapore/china-bubble-tea-brand-singapore/ : Chagee price sightings in Singapore; not a current full menu.

P-eatbook PRESS - in-store sightings, indicative. Eatbook (Aug 2024), https://eatbook.sg/?p=168476 : Chagee price sightings in Singapore; not a current full menu, and older than the deck.

I-team-expansion INFORMAL - team research (Oct 2026): Chagee is looking to expand rapidly in Singapore; how many stores it is going for is confidential. Singapore only. Confirm before citing.

ASSUMPTIONS (ASSUMPTIONS.JSON, ILLUSTRATIVE)

Cups per outlet per year: 200000 cups/store/year

Share of the day's cups in each open hour: 0.04, 0.06, 0.1, 0.11, 0.08, 0.08, 0.09, 0.1, 0.1, 0.09, 0.08, 0.07 share of daily cups per open hour (10:00 to 22:00), sums to 1

Busy and event day multipliers: none 1, moderate 1.5, large 2.5 x forecast arrival rate

Busiest hours the multiplier lands on: 2 busiest forecast hours that get the surge

Seconds per cup by hand, per station: ticket 5, cup ice 10, tea milk 15, topping finish 12, shake seal 10, handover 8 seconds of one average worker per cup

Seconds per cup by robot, per station: ticket 2, cup ice 8, tea milk 10, topping finish 15, shake seal 8, handover 6 seconds per cup per machine unit

Stations the robots do today: ticket, tea milk, shake seal

Spread of worker speeds: 0.2 +/- share around average speed

Minute-to-minute variation in pace: 0.25 coefficient of variation, minute to minute

Share of new staff: 0.25 share of roster in first months

New staff speed: 0.7 x average speed

Coordination loss, people around robots: 0.9 multiplier on human-pool throughput in the mixed setup

Minimum people on the counter: all human 2, mixed 1, all machine 0 workers per open hour

Roster safety margin over the forecast: 1.2 x forecast peak rate

Labour cost per hour worked (SGD): 15.06 SGD per hour worked, fully loaded (statutory costs)

Robot cost per station unit per month (SGD): 1908.75 SGD per station machine unit per month

Attendant hours, all-robot counter: 1 workers per open hour

Walk-away limit for the GM's split table: 0.02 share of peak-hour orders that walk away

Synthetic app-intent dots (slide 05): 3000 synthetic app-intent dots on the map

Walking distance to collect: 1 km a customer will walk to collect

Distance decay, share of the walking distance: 1 distance decay length as a share of walk_km

Pull of not buying: 0.3 pull of not buying, against a store next door (= 1)

DBSCAN cluster radius: 0.6 km

DBSCAN minimum searches: 15 dots within eps to form a cluster core

Map scale: 0.033 km per map pixel

Wait at which a customer walks away: 5 minutes

Price per cup, in store (slide 10): 5.9 SGD per cup, in store

Margin per cup, share of price (slide 10): 0.6 share of price (contribution per avoided order: price minus ingredients and packaging)

Busy days a year (slide 10, illustrative): 104 busy days a year (day type 'moderate')

Event days a year (slide 10, illustrative): 11 event days a year (day type 'large')

Walking-back speed (Week 5 units): 6 lines per hour

CLAIMS WE DID NOT MAKE

No Chagee sales, staffing or wait-time data is used; the forecast is not a trained model. We do not claim the bucket brigade always beats fixed stations: with spare staff, a well-assigned fixed line is as good or slightly faster. Machine breakdowns, breaks and order mix are not modelled.

We make no claim about: **Singapore ownership structure** (a Chagee contact says outlets run under one company and one data system, and that the ownership form does not matter here; I-chagee-contact); **delivery profit by channel**; **Chagee's own equipment cost** (not public); **gross margin per cup** (our assumption); **Chagee's current rostering** (unknown; slide 10's counterfactual is ours).

SPEAKER NOTES · SOURCES AND ASSUMPTIONS

Soft copy only.